PFEIFFER VACUUM

	Pa m³/s W	mbar l/s	Torr l/s	atm cm³/s	lusec	sccm	slm	Mol/s
Pa m³/s	1	10	7.5	9.87	$7.5 \cdot 10^{3}$	592	0.592	$4.41 \cdot 10^{-4}$
mbar l/s	0.1	1	0.75	0.987	750	59.2	5.92 10^{-2}	$4.41 \cdot 10^{-5}$
Torr l/s	0.133	1.33	1	1.32	1000	78.9	$7.89 \cdot 10^{-2}$	$5.85 \cdot 10^{-5}$
atm cm³/s	0.101	1.01	0.76	1	760	59.8	5.98 10^{-2}	$4.45 \cdot 10^{-5}$
lusec	$1.33 \cdot 10^{-4}$	$1.33 \cdot 10^{-3}$	10^{-3}	$1.32 \cdot 10^{-3}$	1	$7.89 \cdot 10^{-2}$	$7.89 \cdot 10^{-5}$	$5.86 \cdot 10^{-8}$
sccm	$1.69 \cdot 10^{-3}$	$1.69 \cdot 10^{-2}$	$1.27 \cdot 10^{-2}$	$1.67 \cdot 10^{-2}$	12.7	1	10^{-3}	$7.45 \cdot 10^{-7}$
slm	1.69	16.9	12.7	16.7	$1.27 \cdot 10^{4}$	1000	1	$7.45 \cdot 10^{-4}$
Mol/s	$2.27 \cdot 10^{3}$	$2.27 \cdot 10^{4}$	$1.7 \cdot 10^{4}$	$2.24 \cdot 10^{4}$	$1.7 \cdot 10^{7}$	$1.34 \cdot 10^{6}$	$1.34 \cdot 10^{3}$	1

	cm	inch	ft		K	C	F
cm	1	0.394	0.033	K	1	K-273.15	9/5 K-459.67
inch	2.54	1	0.083	°C	°C+273.15	1	9/5 °C+32
ft	30.48	12	1	F	5/9 (F+459.67)	5/9 (F-32)	1

°C	F
100	212
80	176
60	140
40	104
20	68
0	32
-20	4
-40	-40

PFEIFFER VACUUM

	Pa N/m²	bar	mbar	µbar dyn/cm²	Torr mm Hg	micron µ, mTorr	atm	at	mm WS	psi lbf/inch²	psf lbf/ft²
Pa	1	$1 \cdot 10^{-5}$	$1 \cdot 10^{-2}$	10	$7.5 \cdot 10^{-3}$	7.5	$9.87 \cdot 10^{-6}$	$1.02 \cdot 10^{-5}$	0.102	$1.45 \cdot 10^{-4}$	$2.09 \cdot 10^{-2}$
bar	$1 \cdot 10^{5}$	1	$1 \cdot 10^{3}$	$1 \cdot 10^{6}$	750	$7.5 \cdot 10^{5}$	0.987	1.02	$1.02 \cdot 10^{4}$	14.5	$2.09 \cdot 10^{3}$
mbar	100	$1 \cdot 10^{-3}$	1	1000	0.75	750	$9.87 \cdot 10^{-4}$	$1.02 \cdot 10^{-3}$	10.2	$1.45 \cdot 10^{-2}$	2.09
µbar	0.1	$1 \cdot 10^{-6}$	$1 \cdot 10^{-3}$	1	$7.5 \cdot 10^{-4}$	0.75	$9.87 \cdot 10^{-7}$	$1.02 \cdot 10^{-6}$	$1.02 \cdot 10^{-2}$	$1.45 \cdot 10^{-5}$	$2.09 \cdot 10^{-3}$
Torr	$1.33 \cdot 10^{2}$	$1.33 \cdot 10^{-3}$	1.33	1330	1	1000	$1.32 \cdot 10^{-3}$	$1.36 \cdot 10^{-3}$	13.6	$1.93 \cdot 10^{-2}$	2.78
micron	0.133	$1.33 \cdot 10^{-6}$	$1.33 \cdot 10^{-3}$	1.33	$1 \cdot 10^{-3}$	1	$1.32 \cdot 10^{-6}$	$1.36 \cdot 10^{-6}$	$1.36 \cdot 10^{-2}$	$1.93 \cdot 10^{-5}$	$2.78 \cdot 10^{-3}$
atm	$1.01 \cdot 10^{5}$	1.013	1013	$1.01 \cdot 10^{6}$	760	$7.6 \cdot 10^{5}$	1	1.03	$1.03 \cdot 10^{4}$	14.7	$2.12 \cdot 10^{3}$
at	$9.81 \cdot 10^{4}$	0.981	981	$9.81 \cdot 10^{5}$	735.6	$7.36 \cdot 10^{5}$	0.968	1	$1 \cdot 10^{4}$	14.2	$2.04 \cdot 10^{3}$
mm WS	9.81	$9.81 \cdot 10^{-5}$	$9.81 \cdot 10^{-2}$	98.1	$7.36 \cdot 10^{-2}$	73.6	$9.68 \cdot 10^{-5}$	$1 \cdot 10^{-4}$	1	$1.42 \cdot 10^{-3}$	0.204
psi	$6.89 \cdot 10^{3}$	$6.89 \cdot 10^{-2}$	68.9	$6.89 \cdot 10^{4}$	51.71	$5.17 \cdot 10^{4}$	6.8 10^{-2}	7.02 10^{-2}	702	1	144
psf	47.8	$4.78 \cdot 10^{-4}$	0.478	478	0.359	359	$4.72 \cdot 10^{-4}$	$4.87 \cdot 10^{-4}$	4.87	$6.94 \cdot 10^{-3}$	1

Nude Hot Cathode Ionization Sensor

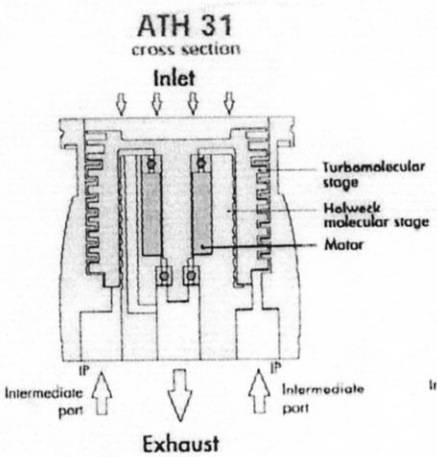
345 Sensor	
Measurement Range	1.0 X 10^{-10} to 1.0 X 10^{-2} Torr 1.3 X 10^{-10} to 1.3 X 10^{-2} mbar 1.3 X 10^{-8} to 1.3 Pascal
Set Point Range	5.0 X 10^{-10} to 9.5 X 10^{-3} Torr 6.5 X 10^{-10} to 1.2 X 10^{-2} mbar 6.5 X 10^{-8} to 1.2 Pascal
Materials Exposed to Vacuum	Tungsten or yttria-coated iridium (filament), nickel, Inconel® X-750, 304 stainless steel, glass, tungsten, platinum
X-Ray Limit	3 X 10^{-10} Torr
Cables and Connectors	Molded tube connector, custom Dsub connector to controller, maximum length is 50 ft
Bakeout Temperature	60°C with cable attached 300°C with CF 150°C with KF
Sensitivity	9 Torr -1 (±20%)
Voltages	Filament bias: +30 V Grid: +180 V Filament: +4.7 V at 1.8 A
Emission Current	1 mA at < 1 X 10^{-4} Torr, 100 iA at > 1 X 10-4 Torr, regulated to ±3%
Degas Power	20 W
Calibration Gas	Air/Nitrogen
Installation Orientation	Any
Weight	0.9 lb (400 g) with CF

ATH series

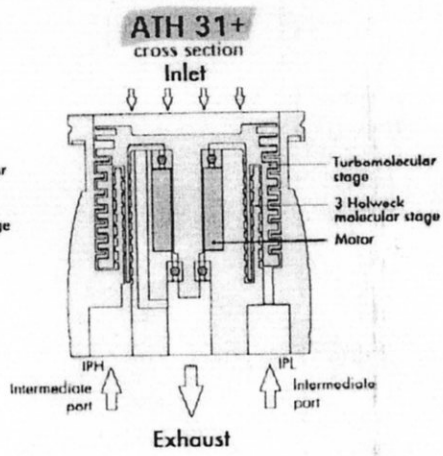
ATH 31: Technical data (measured specifications per PNEUROP standards)

Characteristics		ATH 31			ATH 31+			ATH 31 C		
Inlet flange		Nude	DN 40	DN 63	Nude	DN 40	DN 63	Nude	DN 40	DN 63
Pumping speed	N_2 l/s	33	26	30	33	26	30	33	26	30
	He l/s	15	13	14	22	18	20	22	18	20
	H_2 l/s	7	6	6	15	14	14	15	14	14
Compression ratio	N_2		10^8			10^{11}			2.10^9	
	He		6.10^3			2.10^7			2.10^6	
	H_2		300			1.10^5			1.10^4	
Ultimate pressure (*)	mbar		$< 10^{-8}$			$< 5.10^{-10}$			$< 5.10^{-10}$	
Ultimate pressure with purge	mbar		.			.			1.10^{-8}	
Maximum N_2 purge flow rate	sccm		.			.			25	
Maximum inlet pressure (**)	mbar		1			0.5			1	
Maximum exhaust pressure	mbar		25			45			45	
Recommended fore pump type						RVP, ACP, Membrane Pump				
Mounting orientation						Any				
Rotational speed	rpm					42,000				
Noise level	dBA					45				
Vibration level	mm/s					0.1				
Starting power	VA					100				
Nominal power	W					150				
Start-up time	s					120				
Pump weight	kg (lb)					1.2 (2.7)				
Controllers						ACT 101 H, ACT 201 H				
Cooling						Standard: ambient Forced air cooling: optional Water cooling: optional				

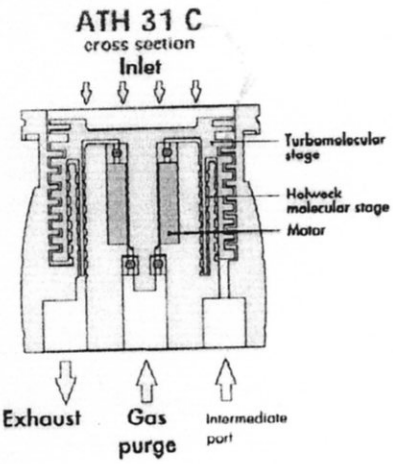
(*): measured following Pneurop standard (CFF flange, after 48 hours of baking with AMD primary pump).
(**): with a 4 m^3/h membrane pump



ATH 31
The ATH 31 includes one turbo stage and one Holweck stage. With excellent compression ratios and a backing pressure tolerance of up to 25 mbar, it can be backed by a membrane pump. It features one intermediate pumping port.



ATH 31+
The ATH 31+, with one turbo stage and three Holweck stages, offers outstanding compression ratios. It is the ideal version for pumping light gases. Its exceptional performance can be achieved with a membrane pump as a fore



ATH 31 C
The ATH 31 C includes one turbo stage, and two Holweck stages. A purge offers protection against corrosive gases. It can be backed by a membrane pump. It features one intermediate pumping port.

ATH series

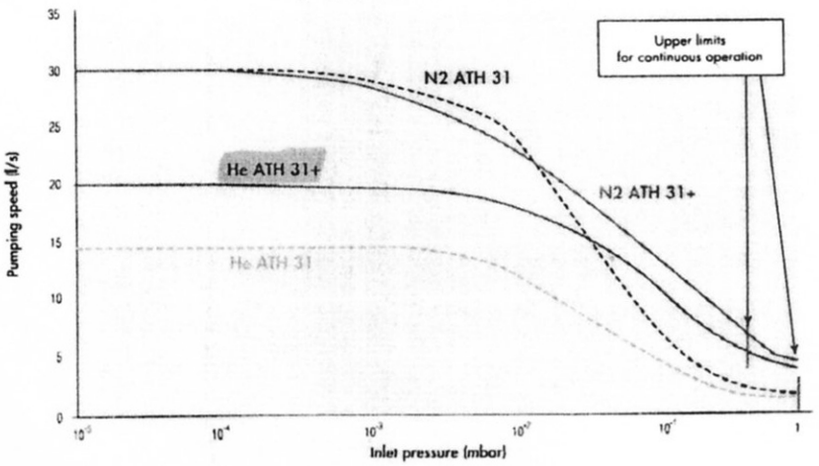
ATH 31

Exceptional performance

An ultimate pressure of 10^{-10} mbar with a dry forepump

The ATH 31+ produces a 10^{11} compression ratio for N_2 and a 10^{5} compression ratio for H_2.
High compression ratios and high backing pressure tolerance (up to 45 mbar) produce very stable performance, ideal for analytical instruments.

Pumping speed versus inlet pressure



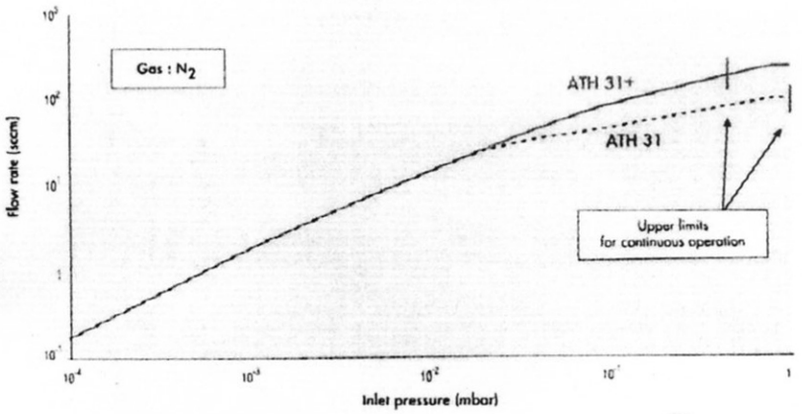
A 200 sccm flow rate at 0.5 mbar

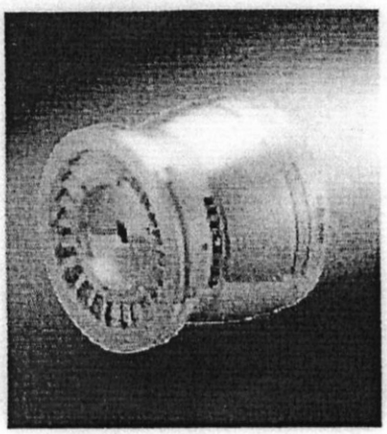
The ATH 31+ allows more than 200 sccm to be pumped continuously



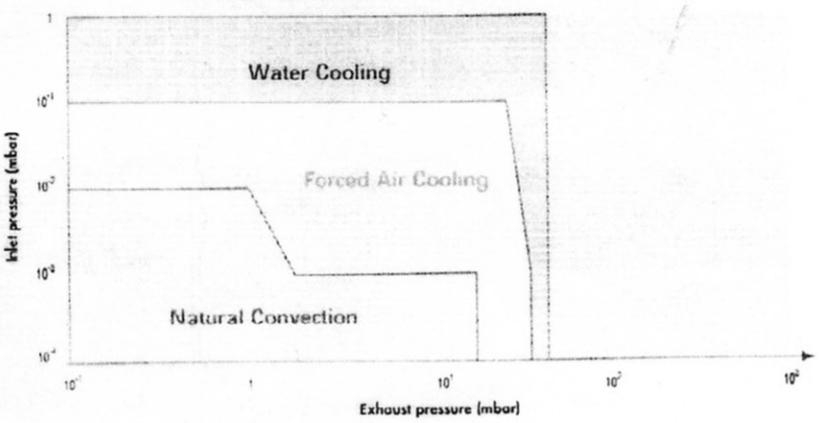
up to 0.5 mbar.

Flow rate versus inlet pressure

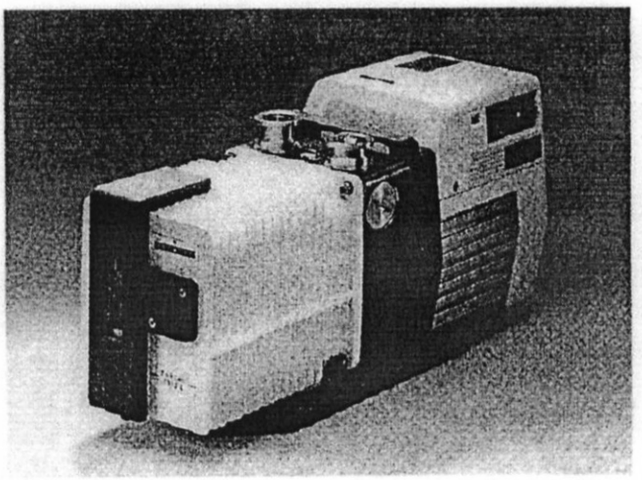


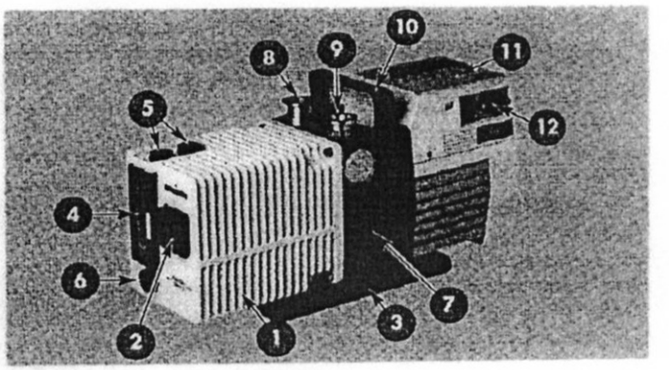


Cooling requirements at (20°C ambient temperature)



Двухступенчатый газоперемещающий механический пластинчато-роторный насос с маслянным уплотнением ALCATEL PASCAL 2005 SD





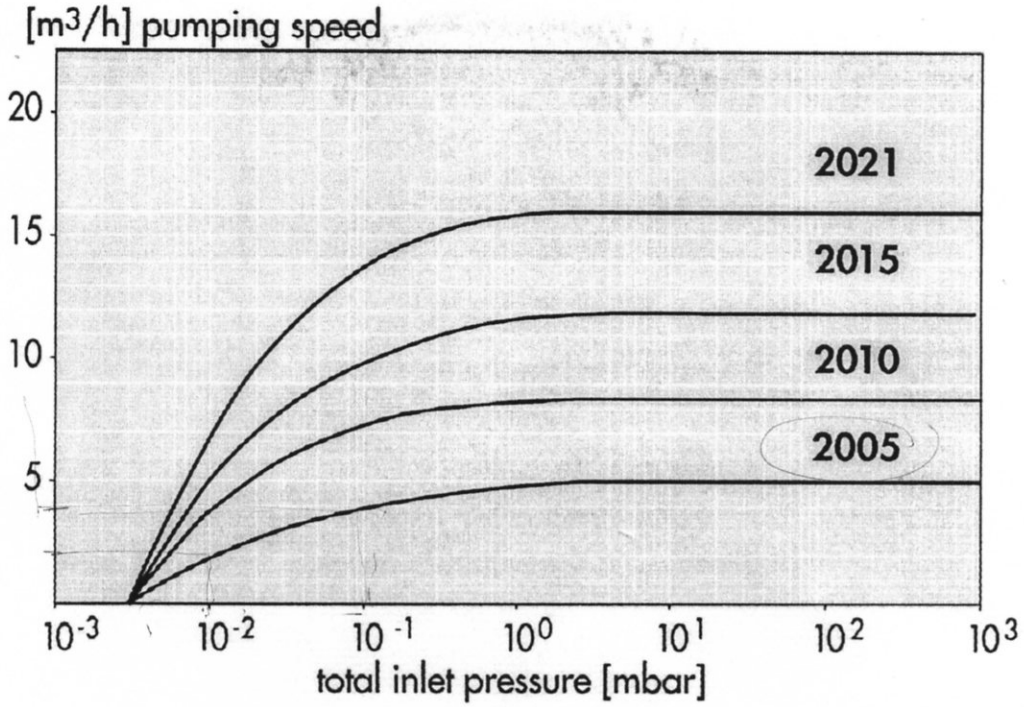
1. Oil case
2. Gas ballast control
3. Base
4. Oil level sight glass
5. Filling plugs
6. Draining plug
7. Frame
8. Inlet end fitting
9. Exhaust end fitting
10. Fold-away handle
11. Electric motor
12. IEC electric socket

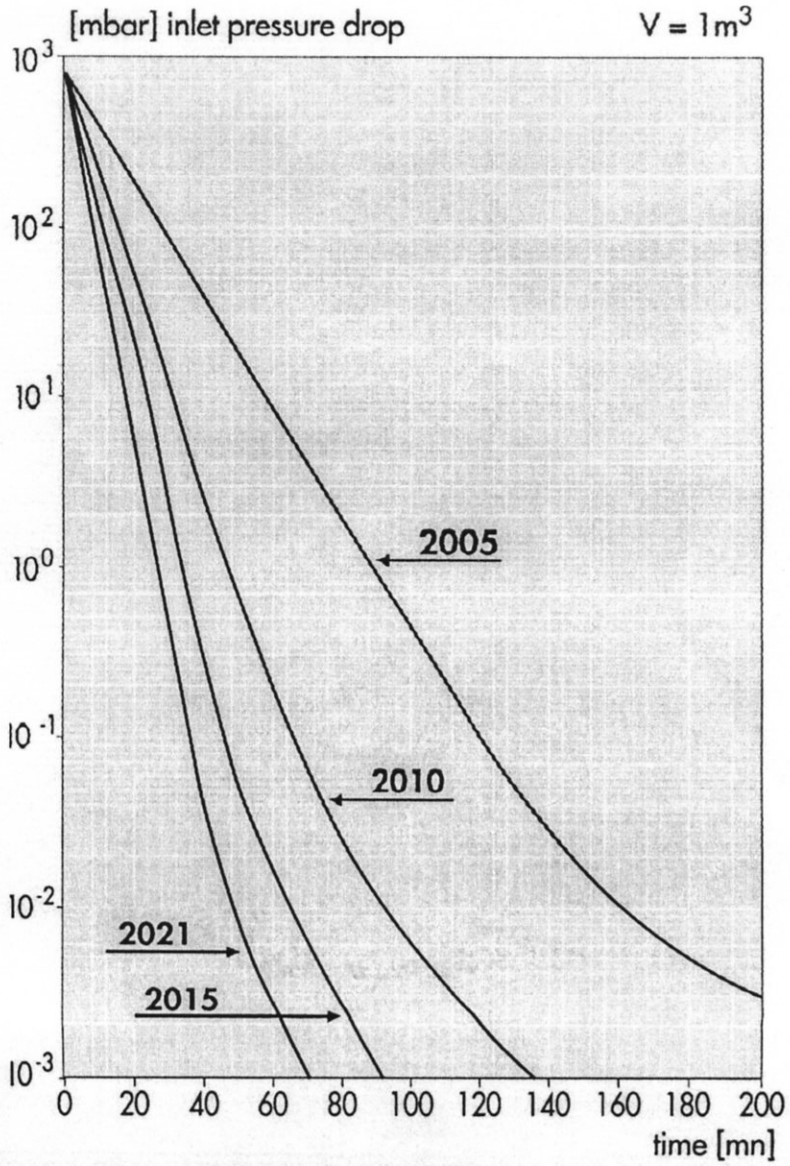
Characteristics	Unit	2005 SD		2010 SD		2015 SD		2021 SD	
Frequency	Hz	50	60	50	60	50	60	50	60
Number of stages		2		2		2		2	
Rotation speed	rpm	1500	1800	1500	1800	1500	1800	1500	1800
Nominal flow rate	m3/h	**5.4**	**6.5**	**9.7**	**11.6**	**15**	**18**	**20.7**	**24.8**
	cfm		3.8		6.8		10.6		14.6
Flow rate Pneurop method	m3/h	**4.8**	**5.7**	**8.5**	**10.2**	**12**	**15**	**15.5**	**20**
	cfm		3.4		6		8.8		11.8
Partial ultimate pressure* with A120 oil	Torr/mbar /Pa	**$7.5.10^{-5}$ / 1.10^{-4}** 1.10^{-2}							
Ultimate pressure with gas ballast closed	Torr/mbar /Pa	**$1.5.10^{-3}$ / 2.10^{-3}** 2.10^{-1}							
Ultimate pressure with gas ballast open	Torr/mbar /Pa	**$7.5.10^{-3}$ / 1.10^{-2}** 1							
Oil capacity	l	0.83		0.950		0.950		0.98	
Weight (pump + motor)**	kg (lbs)	25 (55)		26 (57)		27 (59.5)		28 (62)	
Maximum water vapour pumping capacity (Ballast flowrate 1:1 m^3/h)	mbar	**35**	**25**	**20**	**15**	**12**	**10**	**7**	**7**
	Pa	35.10^2	25.10^2	2.10^3	15.10^2	12.10^2	1.10^3	7.10^2	7.10^2
Water vapour pumping capacity	g/h	120	110	125	100	110	100	90	90
Inlet and exhaust end fittings	ISO-KF	DN 25							

* Partial ultimate pressure measured according to Pneurop 6602 specifications. It may vary if other oils are used ***(See page 69).***

** These values are for pumps equipped with universal single-phase motors.

2005SD - 2010SD
2015SD - 2021SD





Specifications

1179A Mass-Flo®

Analog/Digital, Elastomer-sealed Mass Flow Controller

1179A Mass-Flo®	
Full Scale Ranges	10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000, 20000 sccm (N_2 equivalent)
Maximum Inlet Pressure	150 psig
Normal Operating Pressure Differential (with atmospheric pressure at the MFC outlet)	
10 to 5000 sccm	10 to 40 psid
10000 to 30000 sccm	15 to 40 psid
Control Range	2% to 100% of Full Scale
Accuracy (analog)	±1.0% of Full Scale (including non-linearity, hysteresis, and non-repeatability referenced to 760 mmHg and 0°C)
Repeatability	±0.2% of Full Scale
Resolution	0.1% of Full Scale
Temperature Coefficients	
Zero	<0.05% of Full Scale/°C
Span	<0.08% of Rdg./°C
Warm-up Time	<2 min (to within 0.2% of Full Scale of steady state performance)
Controller Settling Time	<2 sec (per SEMI Guideline E17-91)
Pressure Coefficient	<0.02% of Rdg./psi
Normal Operating Temperature Range	0°C to 50°C
Input Voltage Required	
Max. current at start-up (first 2 sec)	±15 VDC (±5%) @ 200 mA
Typical current at steady state	±15 VDC (±5%) @ 100 mA
Set Point Command Signal	0 to 5 VDC from <20K Ω
Output Signal	0 to 5 VDC into >10K Ω
Output Impedance	<1 Ω
Connector Types	
Analog	9-pin or 15-pin Type "D", 20-pin card edge (The 15-pin Type "D" and card edge connectors are electronically compatible with other MKS flow controllers. Consult Applications Engineering for details.)
Digital	RS-485 DeviceNet™
Wetted Materials	
Standard	316L S.S., Viton®, nickel
Optional (seals and valve seat)	Buna-N, NeopreneR, Kalrez®
Leak Integrity	
External (scc/sec He)	<1 x 10^{-9}
Through closed valve	<1.0% of Full Scale at 40 psig inlet to atmosphere (To assure no flow-through, a separate positive shut-off valve is recommended.)

Specifications

722B Baratron®

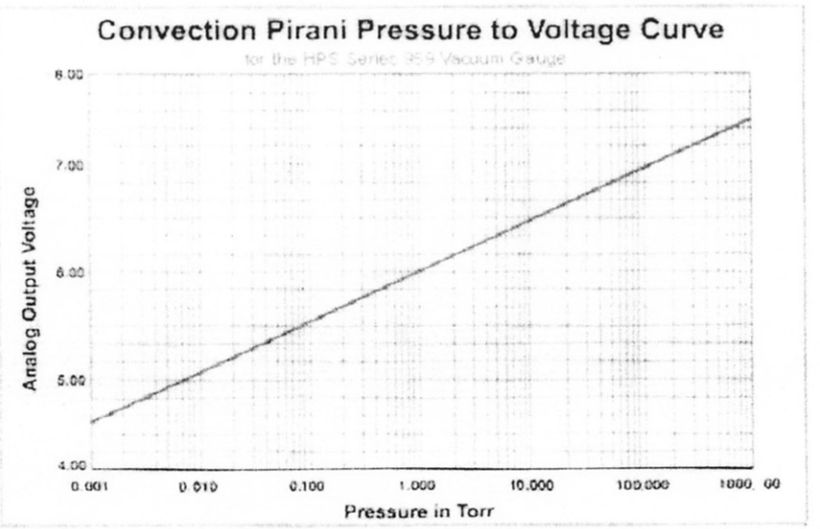
Compact Absolute Capacitance Manometer

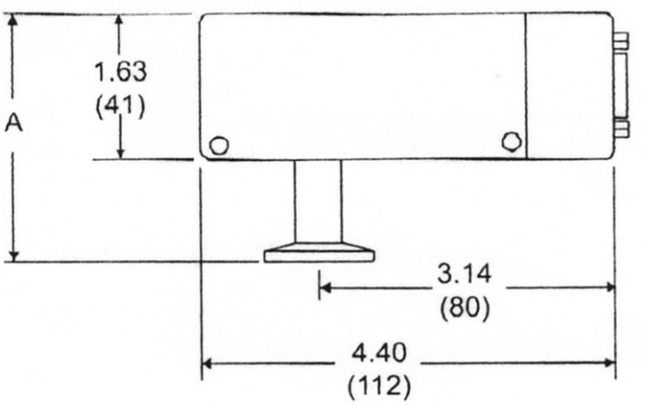
722B Baratron®	
Configuration	Absolute, single-ended
Full Scale Ranges	1 Torr through 25,000 Torr (mmHg) (0.02 psia through 500 psia)
Accuracy	0.5% of Reading (including non-linearity, hysteresis, and non-repeatability)
Response Time	< 20 msec
Temperature Coefficients	
Zero	0.008% of Full Scale/°C (10 Torr through 25,000 Torr); 0.020% of Full Scale (1 and 2 Torr)
Span	0.04% of Reading/°C
Ambient Operating Temperature	0° to 50°C (32° to 122°F), 15° to 40°C (59° to 104°F) (1 and 2 Torr only)
Overpressure Limit	45 psia or 2x Full Scale, whichever is greater
Burst Pressure	10x Full Scale or 100 psi, whichever is greater
Materials Exposed to Gases	Inconel® (Optional Fittings: 316SS)
Power	
Input	0 to 10 Volt output, +13 VDC to +32 VDC @ 10 mA max. (For drop-in replacements of 122A and B which have a ±15 VDC input, use Input/Output Ordering Code 2) 0 to 5 Volt output, +10.8 VDC to +32 VDC @ 10 mA max (regulated if below 13 VDC)
Output	0 to 10 VDC into > 10K ohms load 0 to 5 VDC into > 10K ohms load
Compliance	Fully CE Compliant to EMC Directive 2004/108/EC1 Fully compliant to RoHS Directive 2002-95-EC
Electrical Connectors	9-pin Type "D", 15-pin high density Type "D", 15-pin Type "D" on 4.70 inch (118mm) ±0.5 inch (12.7mm) cable. 4-pin bayonet type, Bendix® - compatible, 5-pin terminal strip
Fittings	
Standard	1/2" tube
Optional	1/4" VCR® female, 8 VCR® female, NW 16 KF2, mini-CF

317

Convection-Enhanced Pirani Vacuum Sensors

317 Sensor	
Measurement Range	1.0 X 10^{-3} to 1.0 X 10^{+3} Torr 1.3 X 10^{-3} to 1.3 X 10^{+3} mbar 1.3 X 10^{-1} to 1.3 X 10^{+5} Pascal
Resolution	1% of indicated decade
Set Point Range	2.0 X 10^{-3} to 9.5 X 10^{+2} Torr 2.7 X 10^{-3} to 1.2 X 10^{+3} mbar 2.7 X 10^{-1} to 1.2 X 10^{+5} Pascal
Reproducibility	5% of indicated pressure
Cables and Connectors	Maximum length is 500 feet 9 pin D-sub connectors, multiconductor shielded cable
Bakeout Temperature	250°C (482°F) with cable and electronics attached
Operating Temperature	0° to 50°C (32° to 122°F)
Materials Exposed to Vacuum	304 stainless steel, nickel 200, glass, platinum
Internal Volume	0.5 $in.^3$ (8.0 cm^3) maximum
Weight	0.5 lb (0.2 kg) 1.34 lb (0.52 kg) with a $2\frac{3}{4}$ inch CF flange





Size	A
NW 16 KF	2.76 (70)